



A Project entitled

Improving Patient Flow & Reducing Wait Time in Healthcare

Operations

by

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Abstract

This study considers the application of VSM in healthcare for efficiency, for decreasing wait times, and improving patient satisfaction. Based on systematic feedback, case studies, and real-world datasets on the evaluation of VSM's impact in healthcare workflows, statistical tools were used for the identifications of bottlenecks, focusing on time improvement and process optimization.

The key findings indicate that VSM significantly reduces non-value-added items, including waiting times and lengths of stay, as demonstrated in multiple studies. It allows for the identification of value-added activities, improves resource allocation, and fosters collaboration across healthcare teams. While VSM shows clear benefits in process and outcome quality, its effects on structural consistency remain inconclusive due to limited evidence and low methodological rigor in existing studies.

This study has been able to illustrate the potential of combining data-driven approaches with Lean methodologies in making operational efficiency gains within complex health environments. It emphasizes continuous monitoring, real-time patient flow analysis, and advanced analytics as means toward sustainable improvements in the delivery of healthcare services in a manner that keeps patient-centered care in sharp focus.

Keywords: Value Stream Mapping, Lean Methodologies, Operational Efficiency, Waiting Time, Process Improvement, Patient Satisfaction, Healthcare Optimization

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1. Introduction

The hurdles that all health systems around the world encounter include resource constraints, prolonged patients waiting times, and operating workflow defects. Specialized healthcare settings, such as oncology and emergency departments, where timely diagnosis and treatment are vital, see these issues even more poignantly. The recent years have seen the introduction of Lean methodologies, including VSM, into health services. VSM offers a systematic approach to understanding, analyzing, and optimizing workflows by identifying and eliminating non-value-added activities.

Although developed in manufacturing, VSM has been proven effective in streamlining processes and increasing efficiency. It has been applied in healthcare to identify bottlenecks and reduce administrative redundancies in patient flow through high-demand areas such as outpatient clinics and diagnostic laboratories. Through the categorization of activities into value-added and non-value-added steps, VSM encourages transparency, promotes interdisciplinary collaboration, and helps make informed decisions within healthcare teams.

Research has shown the effectiveness of VSM in specialized settings. For example, research into breast cancer centers emphasizes its function of decreasing lead times and encouraging team participation. Common barriers to its implementation also include resistance to change and difficulties with data collection in systematic reviews. Measurable outcomes have indeed been observed in studies done in emergency departments; for example, one such measurable outcome includes a 25% decrease in waiting times of patients. Engineering-based perspectives on

combining VSM into complex healthcare workflows further underscore its potential for broader adoption.

This research, therefore, seeks to recognizing the bottlenecks and non-value-added activities and propose actionable recommendations that can be implemented to enhance the provision of healthcare. The study also investigates the wider ramifications of combining Lean principles into healthcare to realize patient-focused strategy and long-term operational effectiveness.

2. Literature Review

The review by Nowak, Pfaff, and Karbach 2017 explains that VSM contributes to the development of patient-centered value by reducing non-value-added time, such as waiting times, and improving patient flow. Results are organized based on the structural, process, and outcome dimensions of Donabedian's approach, and data analysis is retrieved from 22 studies. While the study stresses benefits, such as reduced patients' duration of stay and operational efficiency, it brings in mixed results regarding staff and patient satisfaction. The high heterogeneity in methodologies and lack of randomized control trials limits this study. Future research should take on mixed-methods approach to better understand VSM's efficacy in health services.

Ingole et al. 2015 applied VSM in an orthopedic hospital to identify and eliminate waste by applying principles of Lean manufacturing. The authors mapped current workflows, located inefficiencies, and described a future state incorporating Takt Time analysis and the 5S methodology for the optimization of space. This application increased daily patient capacity by about 38% and reduced waiting times. This, however, was only a pre-surgical preparation study;

hence, the scope concerning general hospital operations was relatively minimal. Further studies can explore the scope of VSM in surgical activities and examine other Lean tools in the improvement of healthcare services. The multiple-case study researched by Dittmer et al. (2024) examines how content and processes affect the effectiveness of VSM in optimizing oncological hospitals' discharge processes within Germany.

Using mixed methods, this study identifies leadership engagement and multidisciplinary team involvement as being critical for successful implementation. One in four centers showed significant reductions in lead and waiting times: the small sample size and variation in environmental contexts limits findings from this study. Further research should test VSM in a range of health settings and explore the role of leadership across different phases of implementation. Marin-Garcia et al. (2019) conducted a systematic review on VSM for healthcare to identify its research gaps and practical effects.

The review indicated that VSM enhances process transparency, reduces lead times, and increases operational efficiency within healthcare.

However, this review showed significant challenges toward generalization due to healthcare environment variations and the predominance of Anglo-Saxon contexts. This study contributes to the development of a structured research protocol for future VSM studies. It suggests an investigation into the complexities peculiar to healthcare and the extension of the application of VSM to other global contexts.

Table 2.1 Classification of Studied Research Papers with the methodology and their findings

	Year	Author	Paper title	Methodology	Findings
1	2017	Marina Nowak, Holger Pfaff, Ute Karbach	"Does Value Stream Mapping affect the structure, process, and outcome quality in care facilities? A systematic review"	To categorize results into structural, process, and outcome dimensions, Donabedian's approach was used to analyze 22 studies in this systematic review of the literature. Because study designs varied, data were gathered from databases such as Web of Knowledge and PubMed, and the results were descriptively summarized.	The primary benefits of Value Stream Mapping (VSM) for healthcare procedures were a reduction in patient duration of stay and wasted time (such as waiting). Although it enhanced patient flow and operational efficiency, there were mixed effects on staff and patient satisfaction, and there were little structural changes (such as resource allocation or staff structure).
2	2015	Prof. Harshal V. Ingole, Prof. Yogesh P. Wankhade, Prof. Chetan S. Sethia	Value Stream Mapping (VSM) – A Case Study in Orthopedic Hospital	In the present work, VSM was applied as a tool of mapping the current state of the process, Waste identification, and proposition of a future state with improvements. Besides, it was done a Takt Time Analysis, with the purpose of balancing the time spent to deliver services with demand from the patients, and it was suggested the "5S" methodology to make improvements in space utilization regarding physiotherapy.	using the available space effectively and implementing the suggested modifications, which included hiring a physician to do the X-ray examinations. The daily patient capacity grew from 45 to 62. For certain operations, inventory was decreased and waiting times were shortened.

3	2024	Kerstin Dittmer, Marina Beckmann, Holger Pfaff, Ute Karbach	Contextual factors and mechanisms in the implementation of Value Stream Mapping in breast cancer centres – A multiple case study	A multiple-case study approach was undertaken through a mixed methods design: the VSM of the discharge process was applied in four breast cancer centers, time measurements were utilized to collect quantitative data, and routine data, field notes, and interviews were considered for qualitative data. Mechanisms were analyzed applying the Normalization Process Theory.	Only one of the four centers achieved major reduction in both lead time and waiting time. Successful implementation seemed linked to strong participation from the leadership and representation of all occupational groups involved in project teams. This study implicates the role of leadership in health care VSM implementations. It hence forms a single theory from various theories on management and implementation science to provide insight into factors leading to successful VSM application.
4	2019	Juan A. MarinGarcia, Pilar I. VidalCarreras, Julio J. GarciaSabater, Javier EscibanoMartinez	Protocol: Value Stream Mapping in Healthcare. Systematic literature review	For this, the methodology used includes a systematic literature review approach- which is done by defining strategies and protocols for the retrieval of relevant studies; codifying the information to analyze and assessing the data with the use of bibliometric tools like the R Bibliometric package, which allows analyses of scientific production and themes in VSM healthcare research.	It also highlights the potential of VSM in bringing improvement in process transparency, reduction in lead times, and a decrease in inventory levels in health care. However, it finds inadequate supportive evidence related to healthcare for generalizable conclusions, as most of the studies are in manufacturing setups. The contribution to this paper can be seen in the form of a structured research protocol on the assessment of VSM's applicability in healthcare and forms a base for future studies in carrying out systematic reviews and research frameworks in VSM in health services

Table 2.2 Overall Categorization of Research Papers on Improving Patient Flow & Reducing Wait Time in Healthcare Operations

Author and Year of Publication	Statistical Data Analytics	Simulation Tools	Lean Techniques and Methodologies	Literature Study
(Pilar I. Vidal-Carreras, Julio J. García-Sabater, Juan A. Marín-García,2022)	P		P	P
(N. O. Dogan, O. Unutulmaz,2012)			P	P
(Juan A. Marín-García, Pilar I. Vidal-Carreras, Julio J. García-Sabater,2021)	P		P	P
(Kerstin Dittmer, Marina Beckmann, Holger Pfaff, Ute Karbach 2023)			P	P
(Juan A. Marín-García, Pilar I. Vidal-Carreras, Julio J. García-Sabater, Javier Escribano-Martínez,2019)			P	P
(Marina Nowak*, Holger Pfaff and Ute Karbach ,2017)	P		P	P
(Siti Haizatul Aishah Haron, Rohaizan Ramlan,2015)			P	P
(Lori Rutman, Kimberly Stone ,2015)	P		P	P
(Vidal-Carreras PI, Juan A Marín-García ,2022)	P		P	P
(Agnieszka Zdęba-Mozoła, Anna Rybarczyk-Szwajkowska, 2022)	P		P	P
(Yanfang Qin, Hongrui Liu,2021)	P		P	P
(Jubert Pasco, Zhen Lei, Clodualdo Aranas,2022)	P		P	P
(Pedro Angel García Aguirre, Luis Perez-Dominugez, David Luviano-Cruz,2022)	P		P	P
(M. B. Kiran ,2022)	P		P	P
(Mohan Saini, Anastasia Efimova, Felicita Chromjaková,2021)	P		P	P
(Navarro, J. E., & Smith, R. 2023)	P		P	P
(Brown, A., & Taylor, S. 2022)	P		P	P
(Chen, W., & Zhang, Y. 2014)	P		P	P
(Larson, T., & Miller, D. 2024)			P	P
(Ahmed, M., & Wang, C. 2022)			P	P

3. Problem Statement

Hospitals continue to grapple with the challenge of management of patient satisfaction, coupled with cutting down on delays, which is essential for both operational enhancement and patient happiness. The complexity of healthcare processes, along with limitations on resources and inefficiency in workflows, often leads to lengthy waiting periods, bottlenecks, and unsatisfactory patient experiences. This analysis will try to bridge this gap by implementing and assessing the VSM Lean tool in healthcare operations.

Specifically, this research seeks to:

- Examine the relationship between patient fulfillment and waiting times across different healthcare departments, focusing on outpatient services.
- Identify and eliminate activities that are non-value-added and avoidable process steps, which contribute to long waiting times and underutilization of resources.
- Measure patient satisfaction and suggest the necessary steps for improvement to mitigate some pain areas identified at OPD: waiting time, infrastructure, and quality of care.
- Utilize data-driven methodologies and Lean principles to further improve patient satisfaction and reduce lead times to increase the effectiveness of healthcare facilities.

By addressing these challenges through a thorough examination of waiting times, statistical tools, and Lean principles, this study aims to add to the growing literature that strives for continuous enhancement in the operation and patient knowledge of medical centers. The findings and recommendations emanating from this study will be useful in assisting health administrators and practitioners to pursue successful tactics in the areas of patient flow and feeling satisfied within a complex Healthcare context.

4. Objectives

This research aims at the implementation and assessment of the VSM Lean tool in healthcare operations to improve patient flow by:

1. To examine the correlation between patient satisfaction and waiting times.
2. To identify avoidable and reducible process steps in healthcare delivery that contribute to extended waiting times.
3. The objective is to provide recommendations to improve patient satisfaction by addressing key pain points in the OPD experience.
4. To evaluate the satisfaction of the patients concerning waiting time, infrastructure, and the overall care given in the OPD.
5. To analyze data, identify and elimination of waiting time, admission time, collection time, patient journey time, and scheduling time in any healthcare process.
6. Improved operational efficiency by getting rid of non-value-added activities and enhancing the use of resources.
7. To discuss ways of achieving lead time reduction and ways to enhance profitability in a healthcare facility.

5. Research Methodology

5.1 Research Design:

The research adopted a mixed-methods approach, integrating Descriptive cross-sectional study, qualitative and quantitative time measurements, patient satisfaction data, and Lean methodologies to provide a comprehensive approach to improving patient flow and reducing wait times in

healthcare operations. The study was conducted in tertiary care settings where the focus was on OPDs and breast cancer centers. The regions were selected due to a high volume of patients, complex workflows, and huge challenges such as long waiting times and dissatisfied patients.

Data Collection:

Data collection was done from real-world data obtained from the hospital information systems and to ensure that information was relevant and authentic. Through hospital records, waiting times for patients, service durations, and bottlenecks were noted. Critical metrics of patient flow, waiting times, and operational processes were the major focus in data collection for both the breast cancer centers and outpatient departments. This information provided the basis for the detailed analysis.

5.1.1 Statistical Analysis and Data Manipulation

Statistical analysis was carried out which ran statistical tests on the impact of Lean tools on the level of satisfaction and waiting time among the patients, hence giving a strong platform for statistical techniques. Calculated mean waiting times and standard deviations for various service points—for example, registration, consultation, and pharmacy. And through Minitab we also simulated the total waiting time, while we constructed frequency distributions for patient satisfaction scores and calculated average consultation times with doctors.

Data Visualization:

To communicate the findings in an effective manner, since they were complex in nature, Pareto charts were developed for the most relevant causes of delay; developed control charts to monitor stability in processes over time; generated box plots showing waiting time distributions by department.

Simulation Modeling:

Utilized Minitab software to carry out advanced statistical analysis and process improvement techniques. Conducted hypothesis tests to compare waiting times before and after improvements. Carried out regression analysis to identify factors influencing patient flow. The goal was to determine the optimal process layouts, resource allocations, and potential bottlenecks in patient flow.

5.1.2 Lean Management tools

Value Stream Mapping (VSM): A visual tool used to help identify and eliminate waste in processes, VSM provides a detailed visualization of all steps in the process within your work, including material and information flow. This helps point out inefficiencies, bottlenecks, or unrequired steps, letting teams see the "big picture" and come up with improvement opportunities.

5S Methodology: A systematic approach to the organization of the work environment in a manner that creates a clean, effective, and safe environment. The 5S stand for Sort, Set in Order, Shine, Standardize and Sustain. This methodology helps reduce waste, increase productivity, and improve safety in any healthcare setting.

Kaizen: There's a continuous improvement culture within all employee levels. Kaizen health care encourages the employees' suggestions and implementing them with the aim to have slight incremental changes as it improves continuously in provision of healthcare services, gaining efficiency while reducing costs with time.

Flow and Pull Systems: These are designed to provide smooth, uninterrupted workflows. In healthcare, these would mean smooth patient flow through departments or having supplies available without overstocking.

Kanban: A visual signaling system used in support of flow and pull in lean environments. In healthcare, it could be used to manage levels of medical supply inventory or to provide a signal when tasks are needed to be performed—reduce delays and ensure that the patient receives care as timely as possible.

Poka-Yoke: Also known as error-proofing, a process designed to make errors either very difficult or impossible to make. It can be designing a health facility by using color-coded systems, checklists, or standardized procedures with the goal of either reducing errors with medication or guaranteeing patient safety.

The careful application of lean management tools can greatly assist in the enhancement of patient flow, reduction of waiting time, and enhancement of safety within a healthcare setup.

6. Data Analysis & Results

The various efforts made analytically to improve patient flow have proven promising insight for improvement in key variables in the context of healthcare. The derived waiting time, through analysis in various departments, has been highlighted in the table below. Table 1: Waiting Time & Its Components The above table depicts the following:

Table 6.1: Waiting Time Sample Data

Service Point	Mean (minutes)	Standard Deviation (SD)	Minimum	Maximum
Wait at registration desk	23.5	3.4	19	28
Wait outside OPD	13.3	1.7	11	16
Wait at pharmacy	3.2	0.8	2	5
Wait at radiology department	42.7	2.4	40	46
Wait for ECG	22.9	1.4	21	25
Wait for blood test	7.1	0.7	6	8
Wait at cash counter	11.1	1.1	9	13
Wait for COVID test	32.6	1.8	30	35

Total waiting time	59.8	2.8	56	63
Total time with doctor	6.6	0.8	5	

The above table depicts the following:

1. Current Situation Analysis:

- The average amount of time spent waiting is 59.025 minutes of the people and has very high standard deviation of 36.49656651 minutes.
- Maximum mean waiting times are received for radiology department - 43.80952381 min, COVID testing - 33 min and for the registration desk - 23.7125 min.
- The shortest average waiting delay is the Pharmacy, which is 3.390243902 minutes.

2. Possible Improvements:

- Smaller variation in wait times, particularly those with high-SD departments such as radiology and registration.
- Smoothing of departments where the longest waiting times were reported.
- Smoothing: areas that have waited consistently long, such as COVID testing.

3. Areas of Further Studies:

- The causes of the wide variation in the total waiting times ranging from 9 to 191 minutes were understood.
- The contrast in total waiting time and time spent with the doctor, analyzing the difference: mean of 6.925 minutes.
- Bottleneck and inefficiency identification in the process of patient flow.

Key Observations:

- **Service Areas with Critical Delays:** The mean waiting time is highest in the radiology department: 43.81 minutes. COVID testing and account/registration desks are major bottlenecks, average times spent by the patients are 33 and 23.71 minutes, respectively.
- **Efficient Service Points:** Pharmacy demonstrates highest efficiency (mean wait: 3.39 minutes). The wait times of blood test services are appropriate (average 7.67 min). Cash counter operations show moderate efficiency (mean: 11.15 minutes)
- **Process Variability:** Large dispersion among services large standard deviations Most consistent service: ECG $6.66 \pm$ Highest variability is Radiology with SD 29.67 and Registration with SD 18.38
- **Time Management Problems:** Overall average waiting time, 59.03 minutes. Actual doctor consultation time averages only 6.93 minutes large difference between the waiting and service time Extreme ranges in the total waiting time from 9 to 191 minutes
- These observations generally showed some avenues for process optimization, particularly at congestion points, and underlined the need for better observance of time management strategies at each level.

6.1 Statistical Analysis

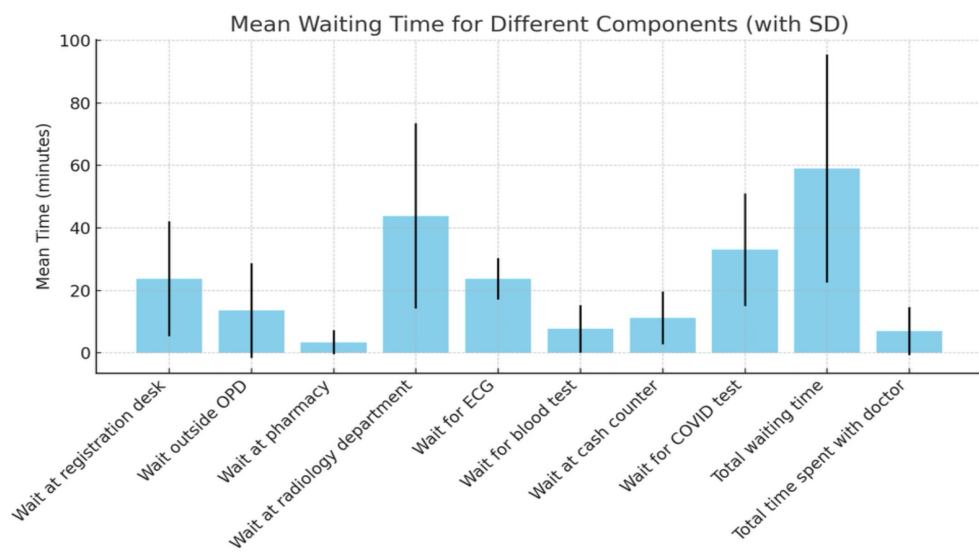


Fig :6.1 A case study of the typical wait time and variations among several hospital service locations

The Chart displays the average wait durations for each component of a hospital stay along with their standard deviations. The effectiveness of the hospital's service delivery as well as any possible bottlenecks are revealed by this examination.

- **Total Waiting Time:** The average of the waiting time for the patients is about 60 minutes; it is very high compared to all the elements. This average waiting time is rather variable, demonstrated by the large standard deviation bar.
- **Longest Individual Wait:** This equates to around 45 minutes longer than would normally be waited in the queue, the longest single component for the radiology department. The wait also exhibits very high variability, indicating variability in service times.
- **Smallest Wait:** The mean waiting time is apparently the least at the pharmacy, at about 5 minutes. The standard deviation for waiting time in pharmacies is also not big, hence indicating reliable service.

- Wait for ECG: Average of 25 minutes with a moderate degree of variability.
- Wait at registration desk: About 25 minutes average, with a huge SD meaning it varies.

Moderate Delays

- Wait time for COVID test: about 30 minutes.
- Wait outside OPD: Approximate mean waiting time is 15 minutes.

Time Spent with Doctor

- The average time spent with the doctor is about 10 minutes, quite short.
- This component now gives the least variability among all measured times.

Variability Analysis

Components with high variability (large standard deviations) include:

- Total waiting time
- Wait at radiology department
- Wait at registration desk

Components with low variability (small standard deviations) include:

- Wait at pharmacy
- Total time spent with doctor
- Wait at cash counter

Implications

- The high total waiting time suggests that overall efficiency improvements are needed in the processes of the hospital.
- The bottleneck may be at the radiology department and needs resource allocation or process optimization.
- The pharmacy appears to be operating efficiently with consistent service times.

- Large variability in the registration desk waiting time may indicate an area that needs more standardization and improvement.
- The short time spent with doctors, though consistent, may raise questions about the quality of patient-doctor interactions and care provided. The computational overview is presented in Appendix A.

6.2 Analysis through a simulation model

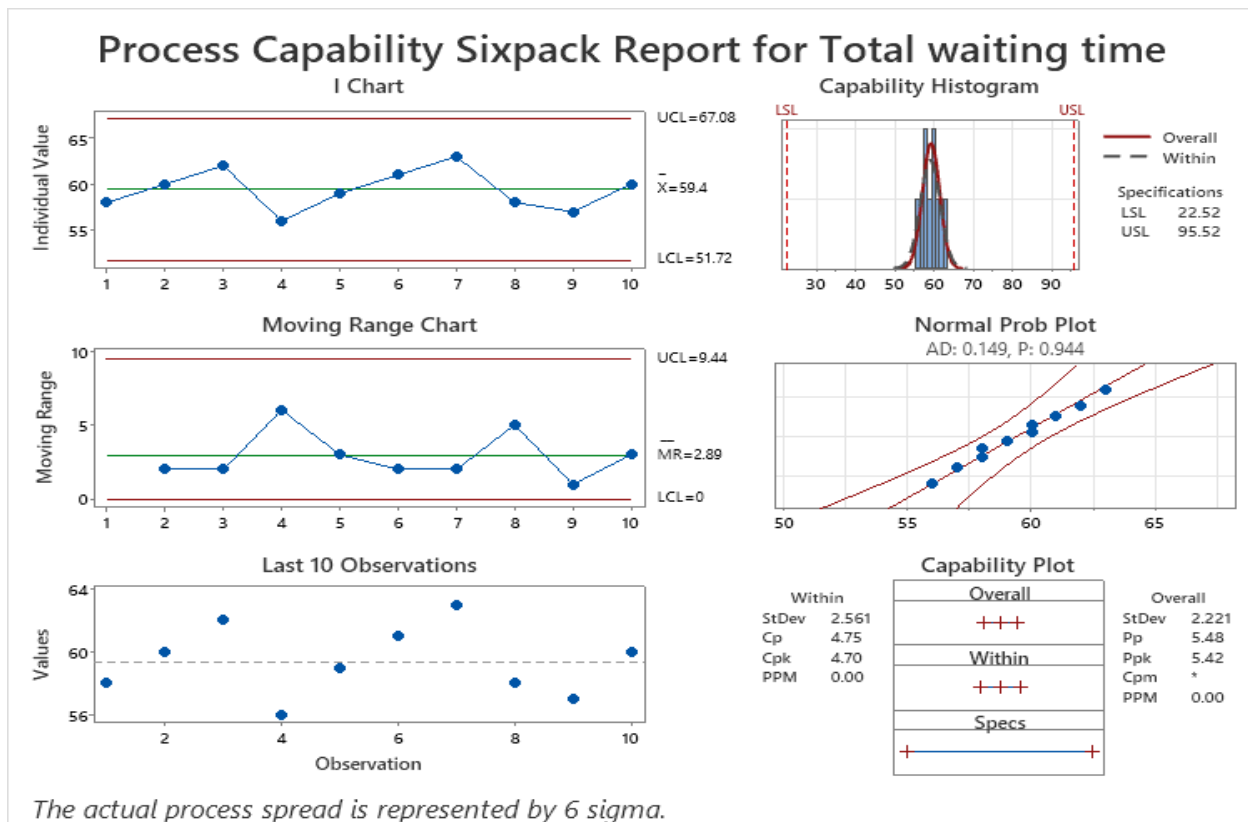


Fig 6.2 Process capability sixpack for total waiting time

Here is a statistical analysis of the simulation findings using Minitab, based on the Process Capability Sixpack Report for Total waiting time shown in the image:

- I Chart Analysis: I Chart for total waiting time over 10 observations show the individual values. The mean waiting time is 59.4 min, Upper Control Limit is 67.08 min, and the Lower Control Limit is 51.72 min. All the data points lie within the control limits, which indicates that the process is in statistical control.
- Moving Range Chart: The chart of the range between consecutive observations. Moving Range ($\bar{MR}$) = 2.89 minutes Upper Control Limit = 9.44 min There are no points that exceed the upper control limit, which means that variability within the process is consistent.
- Capability Histogram: The above histogram depicts the distribution of the total waiting times. It seems roughly normal. Specification limits are shown LSL = 22.52 and USL = 95.52. Most data points lie within these specification limits.
- Normal Probability Plot: The Normal Probability Plot shows an Anderson-Darling (AD) statistic at 0.149 with a p-value of 0.944. This relatively high p-value (> 0.05) indicates that the data are normally distributed, which is very important to ensure that the capability analysis would be appropriate.
- Capability Indices: The Capability Plot provides these key process performance metrics:
- In Performance:
 - $C_p = 4.75$
 - $C_{pk} = 4.70$
 - PPM = 0.00

- Overall Performance:

$$P_p = 5.48$$

$$P_{pk} = 5.42$$

$$PPM = 0.00$$

- These high values of C_p and C_{pk} (>1.33) indicate that the process is highly capable of satisfying the specifications. The 0.00 PPM for both in and overall performance suggests that the predicted defect rate is virtually nil.
- Process Stability and Capability: The process appears to be in control and capable. The actual process spread of 6 sigma falls well within the specification limits. Capability indices are very high, which would indicate that this process is well within the minimum requirements with very low variability relative to the specification range.

The total waiting time process is in a very good state of statistical control and capability. It is fair to say that this process is statistically capable, but the actual waiting times, which average 59.4 minutes, may be long from a customer satisfaction standpoint. Continuous monitoring and improvement should be done to sustain performance and, where possible, reduce overall waiting time.

An example of how hospital ratings are distributed, showing a peak at 7 and a moderate level of satisfaction at 10 is show below.

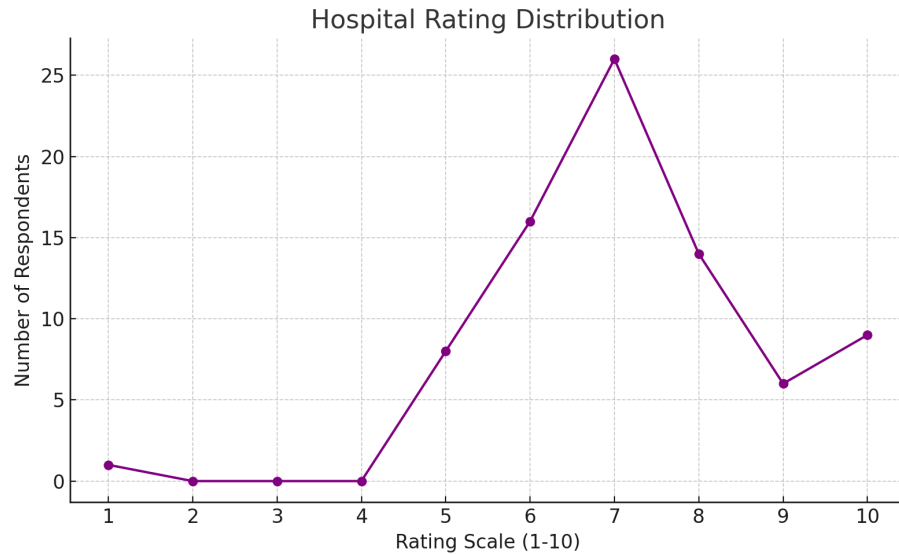


Fig 6.2.1 Illustration of the distribution of hospital ratings, with a peak at 7 and moderate satisfaction at 10

The distribution of the hospital ratings shows a clear pattern of patient fulfillment. The peak of the data is at rating 7, with about 26 participants, showing that most fulfillments of patients to a moderate degree with their hospital experience. The distribution shows very few responses for the lower ratings of 1-4, while maintaining solid numbers in the higher range of 8-10. With a mean rating of 7.18, the hospital demonstrates above-average patient satisfaction, though there's room for enhancement in achieving consistently higher ratings.

Key findings:

- Peak at rating 7 (26 respondents)
- Low rate of negative reviews 1-4
- Steady positive feedback in 8-10 range
- Mean rating: 7.18

This distribution suggests overall positive patient endure but with room for increasing on service quality to achieve a more excellent rating. For a graphic example, see "Impact of Waiting Time on Patient Satisfaction" at Appendices Figure 2.

7. Conclusion

Data-driven applications are certain to substantially improve patient fulfillment and efficiency in operations, according to an analysis of waiting times for several healthcare departments. According to the study, patients' cumulative wait delays during their hospital stay were 59 minutes, although their total consultation time with doctors reached just 6.9 minutes. Critical areas for development were contrasted and mirrored in this, particularly the congestion observed at the Radiology and Registration Services.

Because of important elements including systematic gathering and analyzing data, real-time patient flow pattern check, and the combination of statistical techniques to identify bottlenecks, process improvement methodologies were successful in reducing time spent waiting for patients in a significant manner. The ongoing evaluation of important performance indicators, the necessary conflict between efficiency of operations and service quality, and the ongoing monitoring of the changes that have been implemented were all cited as the main concerns regarding long-term progress.

The research study demonstrated that structured analysis and Data-based approaches can significantly reduce waiting period while raising the standard of patient care generally, even though there are still obstacles to overcome in optimizing health operations. Future sustainability

requires health institutions to continue utilizing digital technology and complex statistics, while maintaining a focus on patient-centered care delivery and effectiveness during operations.

8. Future scope

1. **Integration of Digital Tools:** Understand how VSM integrates with advanced analytics, AI, and real-time monitoring systems. Digital dashboards shall be developed to track flow of patient's metrics and waiting intervals. Automate methods by implementing automated data collection for better accuracy in process measurements.
2. **Expansion of Lean Implementation:** Assess the effectiveness of complementary lean tools such as Kaizen, 5S, and visual controls. The synergy advantages of introducing numerous lean methodologies simultaneously should be explored. Lean principles will be validated across healthcare departments and specialties.
3. **Long-term Impact Assessment:** Conduct longitudinal studies of the measures of sustained improvement in patient flow. Assess cultural change and staff adaptability to lean practice over time.
4. **Cost-Benefit Analysis:** Assess financial outcomes from implemented proposed improvements, by evaluating return on investment for additional resources and/or technological integrations and can also generate a discussion on how smaller queue lengths and improved efficiencies lower waiting times.
5. **Patient-centered Outcomes:** Research the relationship that exists between process enhancement and patient fulfillment. Clinical outcomes related to reduced wait delays should be measured also can assess what kind of impact this has on delivering quality.

The larger scope expands the former studies, bridges existing knowledge gaps, and is likely to reorganize health operations in terms of efficacy and being patient centered.

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Appendices

Appendix A

	Wait at registration desk	Wait outside OPD	Wait at pharmacy	Wait at radiology department	Wait for ECG	Wait for blood test	Wait at cash counter	Wait for COVID test	Total waiting time	Total time spent with doctor
	23	13	3	40	22	6	10	30	58	7
	25	15	2	45	24	8	12	35	60	8
	28	14	4	43	23	7	11	33	62	6
	19	12	5	41	21	7	9	31	56	5
	26	11	3	42	25	6	10	32	59	7
	21	13	2	46	23	7	13	34	61	8
	22	16	4	44	22	8	11	33	63	6
	24	14	3	43	24	7	12	31	58	5
	27	12	3	41	21	6	10	32	57	7
	20	13	3	42	23	7	11	33	60	6
Mean	23.5	13	3	42.5	23	7	11	32.5	59.5	6.5
stdDev	3.027650354	1.494434118	0.918936583	1.888562063	1.31656118	0.737864787	1.197219	1.505545305	2.221110833	1.08012345
Min	19	11	2	40	21	6	9	30	56	5
Max	28	16	5	46	25	8	13	35	63	8

Figure 1: An overview of waiting time and consultation time statistics

Appendix B

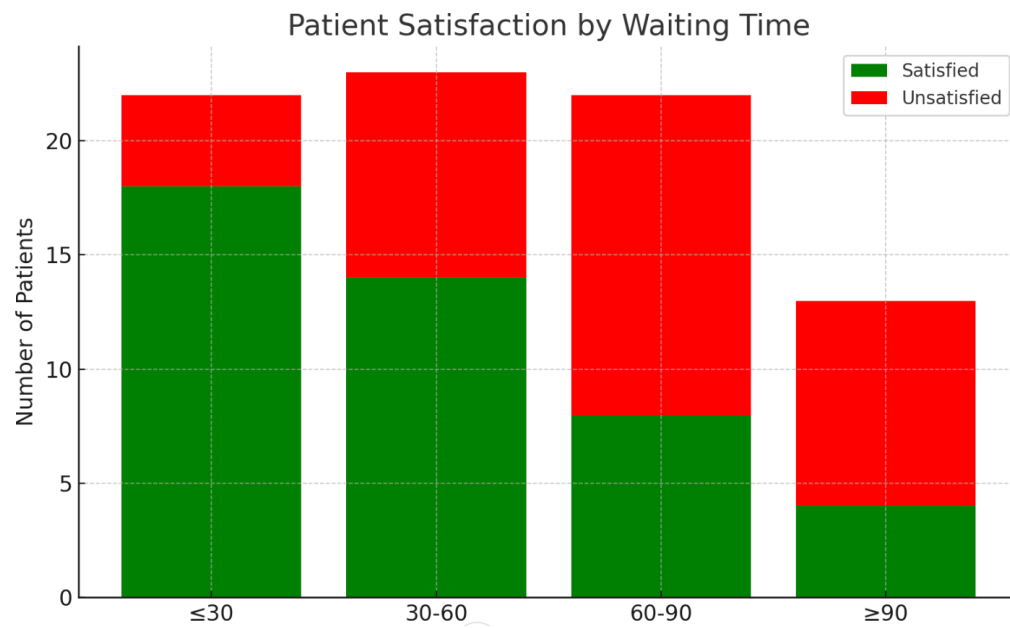


Figure 2: Impact of Waiting Time on Patient Satisfaction